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AP PHYSICS 1

Course Content

The AP Physics 1 course framework provides a clear detailed description of the course requirements for student success. The framework specifies what students must know, be able to do, and understand with a focus on ideals that encompass core principles, theories, and processes of physics. This framework also encourages instruction that prepares students to make connections across domains through a broader way of thinking about the physical world.

UNITS

The course content is organized into commonly taught units. The units have been arranged in a logical sequence frequently found in many college courses and textbooks.

The eight units in AP Physics 1 and their relevant weightings on the multiple-choice section of the AP Exam are listed on the next page.

Pacing recommendations on the Course at a Glance page provide suggestions for how teachers can cover both the required course content and the Progress Checks. The number of suggested class periods is based on a schedule in which the class meets five days a week for 45 minutes each day or for 90 minutes a day for a single semester. While these recommendations have been made to aid in planning, teachers are free to adjust the pacing based on the needs of their students, alternate schedules (e.g., block scheduling), or their school's academic calendar.

Exam Weighting for the Multiple-Choice Section of the AP Physics 1 Exam

Units of Instruction	Exam Weighting
Unit 1: *Kinematics*	**10–15%**
Unit 2: *Force and Translational Dynamics*	**18–23%**
Unit 3: *Work, Energy, and Power*	**18–23%**
Unit 4: *Linear Momentum*	**10–15%**
Unit 5: *Torque and Rotational Dynamics*	**10–15%**
Unit 6: *Energy and Momentum of Rotating Systems*	**5–8%**
Unit 7: *Oscillations*	**5–8%**
Unit 8: *Fluids*	**10–15%**

TOPICS

Each unit is divided into teachable segments called topics. Visit the topic pages (starting on page 31) to see all the required content for each topic.

Learning Objectives and Science Practices

In the AP Physics 1 course and exam, every exam question of student proficiency will be aligned to a learning objective and a skill. The learning objectives represent the content domain, while the skill represents the science practice required to successfully complete the task. The three categories of science practices are described as discrete practices; they are in fact interrelated. For example, scientific questions and predictions are associated with underlying mathematical relationships, and those relationships are used to create diagrams and graphs. The ordering of the science practices is not meant to describe any hierarchy of importance or difficulty.

The three science practices, and their associated skills, will be applied to all learning objectives in the course framework. The task verb "describe," which is used in nearly all learning objectives, encompasses the range of possible graphical, mathematical, or verbal skill applications. Within these multiple representations, students should be able to "describe" a physical concept graphically, mathematically, and verbally.

For example, for a given learning objective, teachers are encouraged to ask the following questions about a physical phenomenon:

- How would students create or interpret graphs or other visual representations?
- What quantitative problems could students solve?
- What experiment could a student design, or what data would students analyze?
- How could the concepts be described verbally?
- How could the course content be used as evidence to justify or support a claim about the behavior of a system, physical phenomena, or outcome of an experiment?

Required Equations

Not all equations in this course framework appear on the equation sheet provided to students while taking the AP Physics 1 Exam. Many of the equations in this document are provided for reference and guidance, or to demonstrate the final results of derivations expected of students on the exam. These equations are denoted as "Derived Equations." Additionally, variables used within this course framework follow the definitions given on the equation sheet. For a complete list of the equations available to students on the AP Physics 1 Exam, please see the AP Physics 1 Table of Information: Equations in the **Appendix**.

Course at a Glance

Plan

The Course at a Glance provides a useful visual organization for the AP Physics 1 course components, including:

- Sequence of units, along with approximate weighting and suggested pacing. Please note, pacing is based on 45-minute class periods, meeting five days each week for a full academic year.
- Progression of topics within each unit.
- Spiraling of the science practices across units.

Teach

PRACTICES
Science Practices spiral throughout the course

1 Creating Representations
2 Mathematical Routines
3 Scientific Questioning and Argumentation

Required Course Content

Each topic contains required Learning Objectives and Essential Knowledge Statements that form the basis of the assessment on the AP Exam.

Assess

Assign the Progress Checks—either as homework or in class—for each unit. Each Progress Check contains formative multiple-choice and free-response questions. The feedback from these checks shows students the areas where they need to focus.

UNIT 1 Kinematics

~12–17 Class Periods | 10–15% AP Exam Weighting

Practices	Topic
1 2 3	1.1 Scalars and Vectors in One Dimension
1 2 3	1.2 Displacement, Velocity, and Acceleration
1 2 3	1.3 Representing Motion
1 2 3	1.4 Reference Frames and Relative Motion
1 2 3	1.5 Vectors and Motion in Two Dimensions

Progress Check 1

Multiple-choice: ~18 questions
Free-response: 4 questions

- Mathematical Routines
- Translation Between Representations
- Experimental Design and Analysis
- Qualitative/Quantitative Translation

UNIT 2 Force and Translational Dynamics

~22–27 Class Periods | 18–23% AP Exam Weighting

Practices	Topic
1 2 3	2.1 Systems and Center of Mass
1 2 3	2.2 Forces and Free-Body Diagrams
1 2 3	2.3 Newton's Third Law
1 2 3	2.4 Newton's First Law
1 2 3	2.5 Newton's Second Law
1 2 3	2.6 Gravitational Force
1 2 3	2.7 Kinetic and Static Friction
1 2 3	2.8 Spring Forces
1 2 3	2.9 Circular Motion

Progress Check 2

Multiple-choice: ~30 questions
Free-response: 4 questions

- Mathematical Routines
- Translation Between Representations
- Experimental Design and Analysis
- Qualitative/Quantitative Translation

UNIT 3 Work, Energy, and Power

~22–27 Class Periods | 18–23% AP Exam Weighting

- 1 2 3 — **3.1** Translational Kinetic Energy
- 1 2 3 — **3.2** Work
- 1 2 3 — **3.3** Potential Energy
- 1 2 3 — **3.4** Conservation of Energy
- 1 2 3 — **3.5** Power

Progress Check 3

Multiple-choice: ~18 questions
Free-response: 4 questions
- Mathematical Routines
- Translation Between Representations
- Experimental Design and Analysis
- Qualitative/Quantitative Translation

UNIT 4 Linear Momentum

~10–15 Class Periods | 10–15% AP Exam Weighting

- 1 2 3 — **4.1** Linear Momentum
- 1 2 3 — **4.2** Change in Momentum and Impulse
- 1 2 3 — **4.3** Conservation of Linear Momentum
- 1 2 3 — **4.4** Elastic and Inelastic Collisions

Progress Check 4

Multiple-choice: ~18 questions
Free-response: 4 questions
- Mathematical Routines
- Translation Between Representations
- Experimental Design and Analysis
- Qualitative/Quantitative Translation

UNIT 5 Torque and Rotational Dynamics

~15–20 Class Periods | 10–15% AP Exam Weighting

- 1 2 3 — **5.1** Rotational Kinematics
- 1 2 3 — **5.2** Connecting Linear and Rotational Motion
- 1 2 3 — **5.3** Torque
- 1 2 3 — **5.4** Rotational Inertia
- 1 2 3 — **5.5** Rotational Equilibrium and Newton's First Law in Rotational Form
- 1 2 3 — **5.6** Newton's Second Law in Rotational Form

Progress Check 5

Multiple-choice: ~18 questions
Free-response: 4 questions
- Mathematical Routines
- Translation Between Representations
- Experimental Design and Analysis
- Qualitative/Quantitative Translation

UNIT 6 Energy and Momentum of Rotating Systems

~8–14 Class Periods | 5–8% AP Exam Weighting

- 1 2 3 **6.1** Rotational Kinetic Energy
- 1 2 3 **6.2** Torque and Work
- 1 2 3 **6.3** Angular Momentum and Angular Impulse
- 1 2 3 **6.4** Conservation of Angular Momentum
- 1 2 3 **6.5** Rolling
- 1 2 3 **6.6** Motion of Orbiting Satellites

Progress Check 6

Multiple-choice: ~18 questions
Free-response: 4 questions

- Mathematical Routines
- Translation Between Representations
- Experimental Design and Analysis
- Qualitative/Quantitative Translation

UNIT 7 Oscillations

~5–10 Class Periods | 5–8% AP Exam Weighting

- 1 2 3 **7.1** Defining Simple Harmonic Motion (SHM)
- 1 2 3 **7.2** Frequency and Period of SHM
- 1 2 3 **7.3** Representing and Analyzing SHM
- 1 2 3 **7.4** Energy of Simple Harmonic Oscillators

Progress Check 7

Multiple-choice: ~18 questions
Free-response: 4 questions

- Mathematical Routines
- Translation Between Representations
- Experimental Design and Analysis
- Qualitative/Quantitative Translation

UNIT 8 Fluids

~12–17 Class Periods | 10–15% AP Exam Weighting

- 1 2 3 **8.1** Internal Structure and Density
- 1 2 3 **8.2** Pressure
- 1 2 3 **8.3** Fluids and Newton's Laws
- 1 2 3 **8.4** Fluids and Conservation Laws

Progress Check 8

Multiple-choice: ~18 questions
Free-response: 4 questions

- Mathematical Routines
- Translation Between Representations
- Experimental Design and Analysis
- Qualitative/Quantitative Translation

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